COMP2001 (AIM) PROJECT REPORT · 2023

Selection Hyper-heuristic for MIN-SET-COVER

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4	2	RL	5
hill-climbing heuristics	ruin-and-recreate heuristics	heuristic selection	best objective value, all trials

1. Solution representation

Representation	Integer array
Alleles	The array length equals the number of subsets in the instance, and each allele represents one subset. For example, instance d1 has 50 subsets, so a solution has 50 alleles (the first subset in d1 contains 97 elements).
Problem instance	A 2D integer array, *arr[totalSets][totalElements]*, stores the elements of each subset as read from the instance file. It is compared against the 1D candidate solution to check that every element is covered.
Initialisation	Random initialisation.

2. Objective function

Nature	Minimising
Function	$f(x_1, \ldots, x_m)$ = minimise |M|, the number of subsets selected.
Infeasible solutions	Not allowed during search. Initial solutions are generated in a loop until a feasible one is found. A *feasible* method returns true only if every element is covered, and a new solution is accepted only if it is both feasible and has a better objective value.

Worked example

Given m = 4, X = {1, 2, 3, 4, 5} and subsets S_1 = {2, 3}, S_2 = {3, 4}, S_3 = {1, 3, 5}, S_4 = {2, 4}:

Candidate solution	Subsets included	Objective value
$\{x_1=1, x_2=1, x_3=0, x_4=1\}$	S_1, S_2, S_4	3
$\{x_1=0, x_2=0, x_3=1, x_4=1\}$	S_3, S_4	2

3. Optimisation approach

3.1 Hyper-heuristic components

Heuristic selection	**Reinforcement learning.** Each heuristic starts with a score of 0. A heuristic gains +1 when it finds an improving solution and loses 1 otherwise. Scores are totalled across iterations, and at the end of a trial the best and worst heuristics are reported and the best is selected.

Move acceptance	**Stochastic, static, improving-only.** The hill climbers propose a new solution each iteration, which is accepted only if it is feasible and has a lower objective value than the best solution so far.

3.2 Low-level heuristics

Heuristic	Category	Param.	Delta eval.
Greedy Heuristic A	Ruin-and-recreate	No	Yes
Greedy Heuristic B	Ruin-and-recreate	No	Yes
First gradient (next descent)	Hill climbing	Yes	Yes
Best gradient (steepest descent)	Hill climbing	No	Yes
Davis's bit hill climbing	Hill climbing	Yes	Yes
Random mutation hill climbing	Hill climbing	Yes	Yes

Ruin-and-recreate	Greedy A deliberately ruins part of the current solution and recreates it by iteratively improving it, helping escape the local optima that pure greedy heuristics get stuck in. Greedy B selects, at each step, the subset that fills the element buffer the most and ignores the rest once the buffer is full. It was added to check whether the hill climbers reach the optimum that this heuristic does.
Parameters	All hill climbers except best gradient take *depth of search* and *intensity of mutation*. Depth of search is the number of iterations a heuristic runs, ending early if a pass finds no improvement. Mutation randomly flips one or more bits. Best gradient is excluded because mutation is more likely to worsen it and extra depth rarely improves a pass.
Not used	No standalone mutation operators and no crossover operators.

3.3 Delta evaluation

The objective value is the number of 1s in the solution. The initial value is stored in a *deltaEv* variable, and each improving bit flip (1 to 0) subtracts 1. This avoids recomputing the objective value of every generated solution, which improves efficiency.

3.4 Termination criterion

A maximum duration in seconds, set by the user. If the user-defined number of steps finishes first, the program reports how long it actually took.

4. Implementation

Code quality	Brief comments and explanations on all classes and methods.
Documentation	A README with instructions for running the code is included.
Compatibility	The code compiles and runs on all provided problem instances.
Output	The code produces the expected output files.

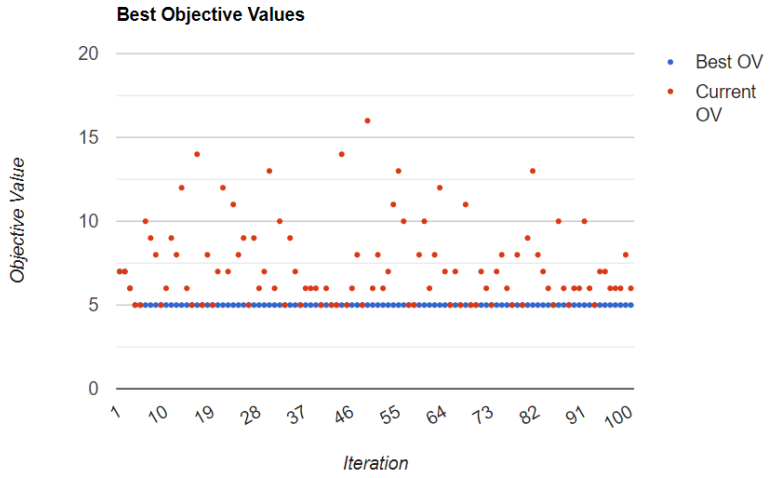
5. Results

Hardware: Intel(R) Core(TM) i5-8265U CPU @ 1.60 GHz (1.80 GHz), Windows 10. Each instance was run for 5 trials of 20 seconds.

Best objective value per trial

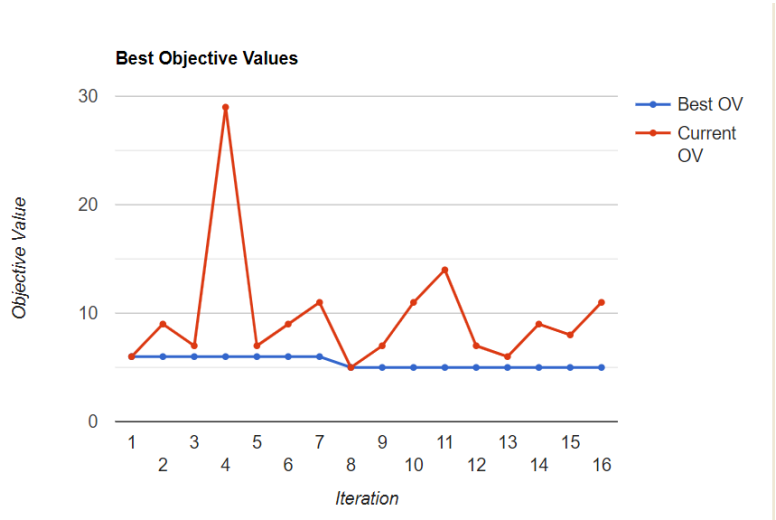
Instance	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
d3_511_210.txt	5	5	5	5	5
d4_2047_495.txt	5	5	5	5	5

d3_511_210.txt, trial 5 (100 steps)



Best and current objective value per iteration.

d4_2047_495.txt, trial 5



Best and current objective value per iteration.

6. Notes

Steepest descent is omitted from the section 5 plots because it performed significantly worse than the other heuristics.